

Crime, geography and policing outcomes: Analysis of rural and urban crime in Kent, UK.

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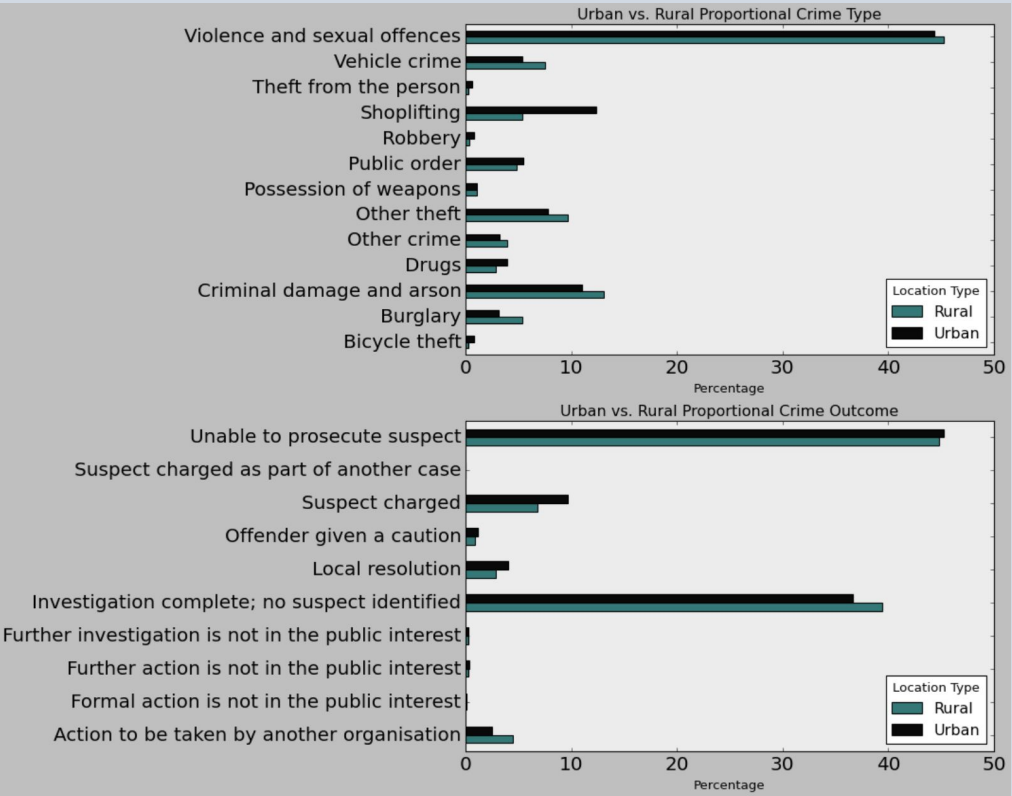
Introduction

Rural criminology suggests that rural communities generally perceive policing as less effective and under-prioritised (Bullock & Garland, 2023). Local policing surveys indicate differing community concerns across Kent (Office of the Kent Police and Crime Commissioner, 2021). In particular, violent offences are prioritised in urban areas, and vehicle crime in rural areas. Using 2024-2025 street-level crime data in Kent as a localised example (Home Office, 2025), this project analysed spatial and temporal features alongside crime types and outcomes to assess literature claims.

Exploratory Analysis

How do the frequency, types, and outcomes of crime differ between rural and urban areas in Kent?

Crime patterns differ between rural and urban areas, although overall distributions remain broadly similar. **Approx. 85% of recorded crimes occurred in urban areas**, although these figures are not adjusted for true population size. **Violence and sexual offences were consistently high in both areas**, accounting for over 40% of crimes. Shoplifting was proportionally higher in urban areas, while vehicle crime, criminal damage and arson, and other theft were more common in rural areas. Crime outcomes were also broadly consistent, with "Unable to prosecute suspect" and "Investigation complete; no suspect identified" accounting for most outcomes. However, **suspect charges were proportionally lower in rural areas**, providing some support for literature suggesting lower perceived policing effectiveness in rural communities.



Preparation

Street-level crime data for Kent was consolidated into a single dataset, undergoing standard processing for null and duplicate values. To obtain an rural/urban spatial flag, the dataset joined with Lower Layer Super Output Area (LSOA) classification data from Office for National Statistics (2021) using LSOA code as the primary key. Irrelevant variables were then removed and those remaining renamed. This supported subsequent exploratory analysis, feature engineering and predictive modelling undertaken to address the research questions.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 285895 entries, 0 to 285894
Data columns (total 7 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Crime ID             285895 non-null object
1   Month                285895 non-null object
2   LSOA code            285895 non-null object
3   Crime type           285895 non-null object
4   Last outcome category 285895 non-null object
5   Outcome type         285895 non-null object
6   Urban_rural_flag     285895 non-null object
dtypes: object(7)
memory usage: 15.3+ MB
```

Dataset structure and basic summary after preprocessing.

Multinomial Logistic Regression

Can crime type be predicted using spatial and temporal features?

Further exploratory analysis suggested temporal features were feasible for modelling due to some quarterly variation in crime type. However, these patterns were inconsistent and indicate limited predictive potential. Class imbalance was addressed by removing minor crime types, although substantial imbalance remained.

Whilst appropriate for the dataset and task, the initial multinomial logistic regression model did not yield meaningful predictive performance for crime type. The first iteration was built using one spatial and temporal feature, achieving an F1-score of 0.09. **This performance is close to random classification.**

A second iteration of the model was run. This introduced additional feature engineering, including a spatial-temporal interaction term and frequency encoded LSOA codes. However, this produced negligible improvement. In the absence of additional insight, it suggests **crime type cannot be reliably predicted using spatial and temporal features alone.**

	precision	recall	f1-score	support
Burglary	0.06	0.21	0.09	1994
Criminal damage and arson	0.00	0.00	0.00	6459
Drugs	0.05	0.13	0.07	2168
Other crime	0.00	0.00	0.00	1924
Other theft	0.09	0.02	0.03	4609
Possession of weapons	0.00	0.00	0.00	631
Public order	0.06	0.25	0.10	3856
Shoplifting	0.14	0.47	0.21	6468
Vehicle crime	0.07	0.13	0.09	3263
Violence and sexual offences	0.00	0.00	0.00	25445
accuracy			0.09	56017
macro avg	0.05	0.12	0.06	56017
weighted avg	0.03	0.09	0.04	56017

Classification report for first iteration of multinomial logistic regression.

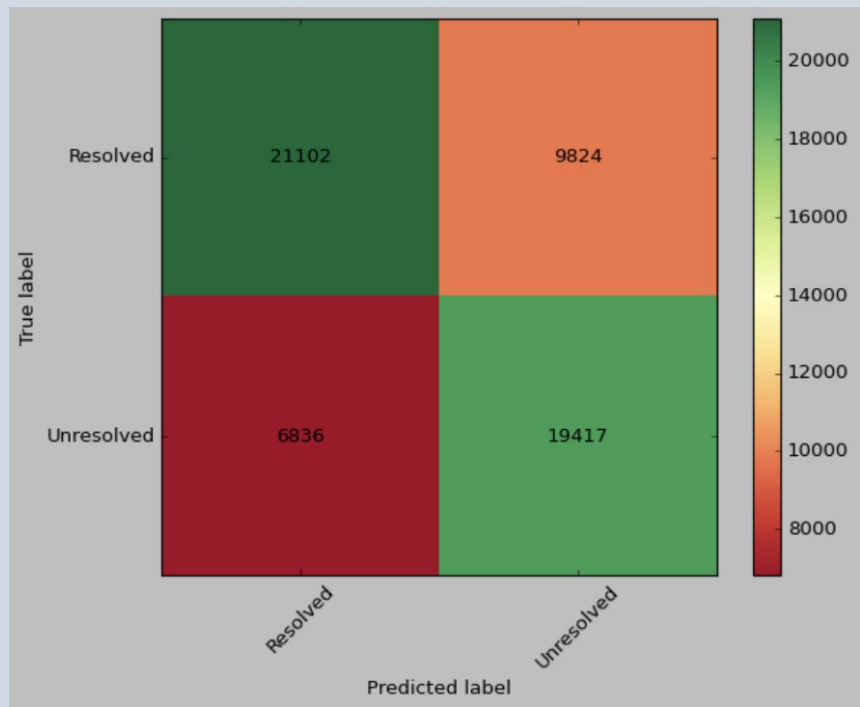
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Public order	0.06	0.25	0.10	3856
Shoplifting	0.14	0.47	0.21	6468
Vehicle crime	0.07	0.18	0.10	3263
Violence and sexual offences	0.00	0.00	0.00	25445
accuracy			0.09	56017
macro avg	0.05	0.12	0.06	56017
weighted avg	0.04	0.09	0.05	56017

Classification report for second iteration after feature engineering.

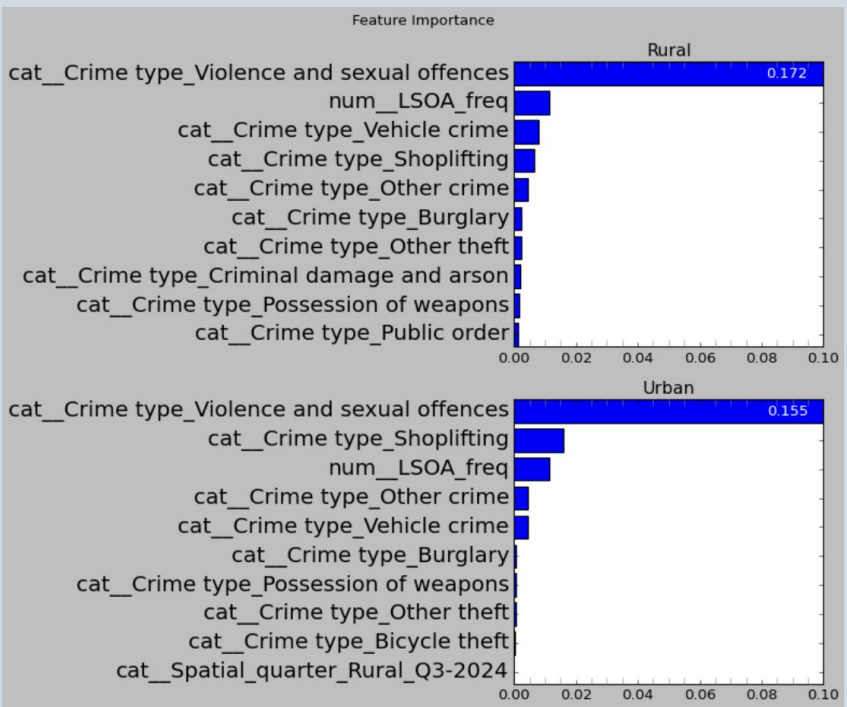
Random Forest Classification

Do predicted crime outcomes differ systematically between rural and urban areas, and do these patterns align with community policing priorities?

The model demonstrated strong performance in predicting binary crime outcomes. Five-fold cross-validation produced a mean F1-score of 0.71, showing good generalisability against near identical test performance. Rural and urban subset performance was also consistent, indicating that **broad spatial attributes do not describe systematic differences in crime outcomes**. However, lower-level geography (LSOA) is more influential. Feature importance analysis highlighted crime type as the dominant predictor of outcomes. Notably, **vehicle crime demonstrated greater predictive importance within rural areas**, providing partial support for reported rural community policing priorities.



Confusion matrix for cross-validated Random Forest classifier.



Feature importance plots under rural/urban permutation importance.